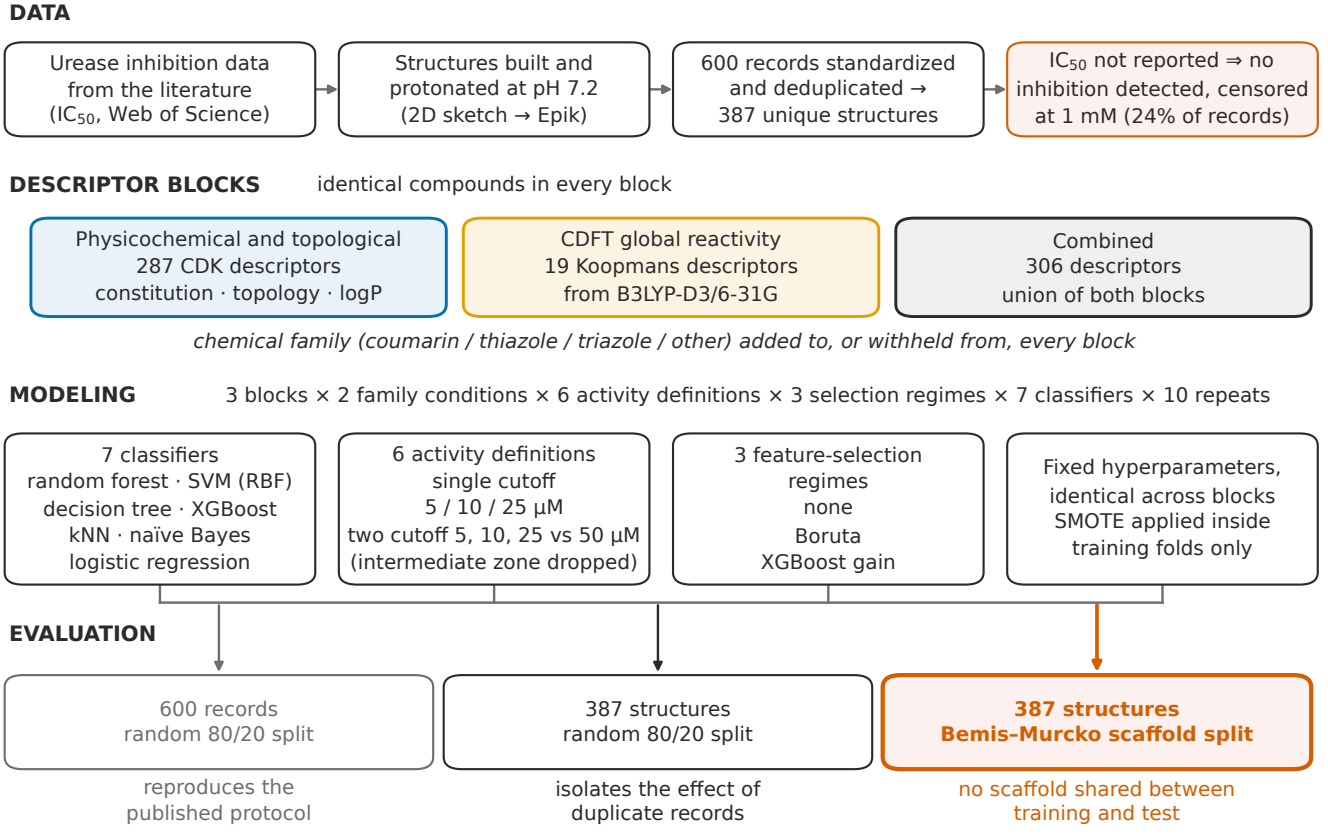
DATA
Urease inhibition data from the literature (IC_{50}, Web of Science)
Structures built and protonated at pH 7.2 (2D sketch → Epik)
600 records standardized and deduplicated → 387 unique structures
IC_{50} not reported ⇒ no inhibition detected, censored at 1 mM (24% of records)
DESCRIPTOR BLOCKS
identical compounds in every block
Physicochemical and topological
287 CDK descriptors
constitution · topology · logP
CDFT global reactivity
19 Koopmans descriptors
from B3LYP-D3/6-31G
Combined
306 descriptors
union of both blocks
chemical family (coumarin / thiazole / triazole / other) added to, or withheld from, every block
MODELING
3 blocks × 2 family conditions × 6 activity definitions × 3 selection regimes × 7 classifiers × 10 repeats
7 classifiers
random forest · SVM (RBF)
decision tree · XGBoost
kNN · naïve Bayes
logistic regression
6 activity definitions
single cutoff
5 / 10 / 25 µM
two cutoff 5, 10, 25 vs 50 µM
(intermediate zone dropped)
3 feature-selection regimes
none
Boruta
XGBoost gain
Fixed hyperparameters, identical across blocks
SMOTE applied inside training folds only
EVALUATION
600 records
random 80/20 split
reproduces the published protocol
387 structures
random 80/20 split
isolates the effect of duplicate records
387 structures
Bemis–Murcko scaffold split
no scaffold shared between training and test
95% confidence intervals over repeats · blocks compared by paired Friedman and Wilcoxon tests on matched configurations · SHAP attribution by descriptor origin